

Exercise 1.

ALL IS WELL IN THE NEWELL FAMILY

WORD	LENGTH (X)	NUMBER OF L's (Y)
ALL	3	2
IS	2	0
WELL	4	2
IN	2	0
THE	3	0
NEWELL	6	2
FAMILY	6	1

(a) probability of selecting any word: $P(W) = \frac{1}{7}$ Joint PMF $P(X = x, Y = y)$:

$X = x$	$Y = y$	$P(X = x, Y = y)$
3	2	1/7
2	0	2/7
4	2	1/7
3	0	1/7
6	2	1/7
6	1	1/7

Marginal PDF of X :

$X = x$	$P(X = x)$
2	2/7
3	2/7
4	1/7
6	2/7

Marginal PDF of Y :

$Y = y$	$P(Y = y)$
0	3/7
1	1/7
2	3/7

(b)

- pay \$1 dollar for each letter in a word
- get \$4 dollars for each L in a word

$$R(X, Y) = 4Y - X$$

$$R(X, Y) \geq 0 \rightarrow 4Y - X \geq 0 \rightarrow 4Y \geq X \rightarrow Y \geq \frac{X}{4}$$

$X = x$	$Y = y$	$4Y - X \geq 0$
3	2	$5 \geq 0$ - yes
2	0	$-2 \geq 0$ - no
4	2	$0 \geq 0$ - yes
2	0	$-2 \geq 0$ - no
3	0	$-3 \geq 0$ - no
6	2	$2 \geq 0$ - yes
6	1	$-2 \geq 0$ - no

$$P(\text{breakeven}) = P(X = 3, Y = 2) + P(X = 4, Y = 2) + P(X = 6, Y = 2)$$

$$= \frac{1}{7} + \frac{1}{7} + \frac{1}{7}$$

$$= \frac{3}{7}$$

(c) "at least one L", $Y \geq 1$, i.e. $Y = 1, Y = 2$

- "ALL" (3,2)
- "WELL" (4,2)
- "NEWELL" (6,2)
- "FAMILY" (6,1)

$$P(Y \geq 1) = P(X = 3, Y = 2) + P(X = 4, Y = 2) + P(X = 6, Y = 2) + P(X = 6, Y = 1)$$

$$= \frac{1}{7} + \frac{1}{7} + \frac{1}{7} + \frac{1}{7}$$

$$= \frac{4}{7}$$

$$P(\text{breakeven and } Y \geq 1) = \frac{3}{7}, (6,1) \text{ doesn't break even}$$

$$P(\text{break} | Y \geq 1) = \frac{P(\text{breakeven and } Y \leq 1)}{P(Y \geq 1)} = \frac{\frac{3}{7}}{\frac{4}{7}} = \frac{3}{4}$$

Exercise 2.

(a) $\Omega = 16$

- $X = \max(\text{first die, second die}), Y = \min(\text{first die, second die})$
- values of X and Y can be 1-4

X=1

$$P(X = 1, Y = 1) = \frac{1}{16}$$

X=2

$$P(X = 2, Y = 1) = \frac{2}{16}$$

$$P(X = 2, Y = 2) = \frac{1}{16}$$

X=3

$$P(X = 3, Y = 1) = \frac{2}{16}$$

$$P(X = 3, Y = 2) = \frac{2}{16}$$

$$P(X = 3, Y = 3) = \frac{1}{16}$$

X=4

$$P(X = 4, Y = 1) = \frac{2}{16}$$

$$P(X = 4, Y = 2) = \frac{2}{16}$$

$$P(X = 4, Y = 3) = \frac{2}{16}$$

$$P(X = 4, Y = 4) = \frac{1}{16}$$

Y/X	1	2	3	4
1	1/16	2/16	2/16	2/16
2	0	1/16	2/16	2/16
3	0	0	1/16	2/16
4	0	0	0	1/16

Marginal PDF of X :

$X = x$	$P(X = x)$
1	1/16
2	3/16
3	5/16
4	7/16

Marginal PDF of Y :

$Y = y$	$P(Y = y)$
1	7/16
2	5/16
3	3/16
4	1/16

(b)

$$E(X) = \sum_{x=1}^4 xP(X = x) = 1 \times \frac{1}{16} + 2 \times \frac{3}{16} + 3 \times \frac{5}{16} + 4 \times \frac{7}{16}$$

$$E(X) = \frac{1}{16} + \frac{6}{16} + \frac{15}{16} + \frac{28}{16}$$

$$E(X) = \frac{50}{16}$$

$$E(Y) = \sum_{y=1}^4 yP(Y = y) = 1 \times \frac{7}{16} + 2 \times \frac{5}{16} + 3 \times \frac{3}{16} + 4 \times \frac{1}{16}$$

$$E(Y) = \frac{7}{16} + \frac{10}{16} + \frac{9}{16} + \frac{4}{16}$$

$$E(Y) = \frac{30}{16}$$

(c) $U = \frac{X}{Y}$

$X = x$	$Y = y$	$U = u$	$P(U = u)$
1	1	1/1 = 1	1/16
2	1	2/1 = 2	2/16
2	2	2/2 = 1	1/16
3	1	3/1 = 3	2/16
3	2	3/2 = 1.5	2/16
3	3	3/3 = 1	1/16
4	1	4/1 = 4	2/16
4	2	4/2 = 2	2/16
4	3	4/3 = 1.3	2/16
4	4	4/4 = 1	1/16

PMF of U :

U	$P(U = u)$
1	4/16
1.33	2/16
1.5	2/16
2	4/16
3	2/16
4	2/16

$$(d) E\left(\frac{X}{Y}\right) = 1 \times \frac{4}{16} + 1.33 \times \frac{2}{16} + 1.5 \times \frac{2}{16} + 2 \times \frac{4}{16} + 3 \times \frac{2}{16} + 4 \times \frac{2}{16}$$

$$E\left(\frac{X}{Y}\right) = \frac{4}{16} + \frac{1}{6} + \frac{3}{16} + \frac{8}{16} + \frac{6}{16} + \frac{8}{16}$$

$$E\left(\frac{X}{Y}\right) = \frac{95}{48} = 1.979166667$$

$$\frac{E(X)}{E(Y)} = \frac{\frac{50}{16}}{\frac{30}{16}} = \frac{50}{30} = 1.666666667$$

Therefore, $E\left(\frac{X}{Y}\right) \neq \frac{E(X)}{E(Y)}$

Exercise 3.

Two random variables are independent if:

$$P(X \leq x, Y \leq y) = P(X \leq x)P(Y \leq y)$$

$$F_X(x) = P(X \leq x)$$

$$F_Y(y) = P(Y \leq y)$$

$$F_{X,Y}(x, y) = F_X(x)F_Y(y)$$

$$f(x, y) = \frac{d^2}{dx dy} F_{X,Y}(x, y)$$

$$f(x, y) = \frac{d^2}{dx dy} (F_X(x)F_Y(y))$$

$$f(x, y) = \frac{d}{dx} F_X(x) \frac{d}{dy} F_Y(y)$$

Marginal PDF X : $f_X(x) = \frac{d}{dx} F_X(x)$

Marginal PDF Y : $f_Y(x) = \frac{d}{dy}F_Y(y)$

$$f(x, y) = f_X(x)f_Y(y)$$

We have shown X and Y are independent, the joint PDF factors as $f(x, y) = f_X(x)f_Y(y)$, which is of the form $f(x, y) = g(x)h(y)$, where $g(x) = f_X(x)$ and $h(y) = f_Y(y)$

$$f(x, y) = g(x)h(y)$$

$$f_X(x) = \int_{-\infty}^{\infty} f(x, y) dy$$

$$= \int_{-\infty}^{\infty} g(x)h(y) dy$$

$$= g(x) \int_{-\infty}^{\infty} h(y) dy$$

Let $c_1 = \int_{-\infty}^{\infty} h(y) dy$, is a constant. Thus, the marginal PDF $f_X(x)$ is: $f_X(x) = g(x)c_1$

$$f_Y(y) = \int_{-\infty}^{\infty} f(x, y) dx$$

$$= \int_{-\infty}^{\infty} g(x)h(y) dx$$

$$= h(y) \int_{-\infty}^{\infty} g(x) dx$$

Let $c_2 = \int_{-\infty}^{\infty} g(x) dx$, is a constant. Thus, the marginal PDF $f_Y(y)$ is: $f_Y(y) = h(y)c_2$

$$F_{X,Y}(x, y) = \int_{-\infty}^x \int_{-\infty}^y f(u, v) dv du$$

$$= \int_{-\infty}^x g(u) du \int_{-\infty}^y h(v) dv$$

The first integral is the marginal CDF $F_X(x)$, and the second integral is the marginal CDF $F_Y(y)$:

$$F_{X,Y}(x, y) = F_X(x)F_Y(y)$$

Since the joint CDF factors as $F_{X,Y}(x, y) = F_X(x)F_Y(y)$, by the definition of independence, X and Y are independent.

Exercise 4.

$$f_{X,Y}(x, y) = \begin{cases} \frac{1}{2}(x+y)e^{-(x+y)}, & x > 0, y > 0 \\ 0, & \text{elsewhere} \end{cases}$$

For $f_{X,Y}(x, y)$ to be a valid joint probability density function (PDF), it must satisfy two conditions:

1. **Non-negativity:** $f_{X,Y}(x, y) \geq 0$ for all x and y
2. **Normalization:** The total probability over the entire space must be 1

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{X,Y}(x, y) dx dy = 1$$

1. Non-negativity

$f_{X,Y}(x, y) = \frac{1}{2}(x+y)e^{-(x+y)}$ is non-negative for $x > 0$ and $y > 0$ because:

- the term $x+y$ is positive for $x > 0$ and $y > 0$
- the exponential function $e^{-(x+y)}$ is always positive

Therefore, $f_{X,Y}(x, y) \geq 0$ for $x > 0$ and $y > 0$, and $f_{X,Y}(x, y) = 0$ elsewhere.

2. Normalization:

$$\begin{aligned} &= \int_0^{\infty} \int_0^{\infty} \frac{1}{2}(x+y)e^{-(x+y)} dx dy \\ &= \frac{1}{2} \left[\int_0^{\infty} \int_0^{\infty} xe^{-(x+y)} dx dy + \int_0^{\infty} \int_0^{\infty} ye^{-(x+y)} dx dy \right] \\ &= \frac{1}{2} \left[\int_0^{\infty} e^{-y} \int_0^{\infty} xe^{-x} dx dy + \int_0^{\infty} ye^{-y} \int_0^{\infty} e^{-x} dx dy \right] \\ &= \frac{1}{2} \left[\int_0^{\infty} e^{-y} (1^2 \Gamma(2)) dy + \int_0^{\infty} ye^{-y} \times 1 dy \right] \\ &= \frac{1}{2} \left[\int_0^{\infty} e^{-y} \times 1 dy + \int_0^{\infty} ye^{-y} dy \right] \\ &= \frac{1}{2} \left[\int_0^{\infty} e^{-y} \times 1 + \int_0^{\infty} ye^{-y} dy \right] \\ &= \frac{1}{2} (1 + (1^2 \Gamma(2))) \\ &= \frac{1}{2} (1 + 1) \\ &= \frac{1}{2} \times 2 \\ &= 1 \end{aligned}$$

Marginal PDF of X:

$$\begin{aligned} f_X(x) &= \int_0^{\infty} \frac{1}{2}(x+y)e^{-(x+y)} dy \\ f_Y(x) &= \frac{1}{2} \left[\int_0^{\infty} xe^{-(x+y)} dy + \int_0^{\infty} ye^{-(x+y)} dy \right] \\ &= \frac{1}{2} e^{-x} \left[\int_0^{\infty} xe^{-y} dy + \int_0^{\infty} ye^{-y} dy \right] \end{aligned}$$

$$= \frac{1}{2}e^{-x} \left[x \int_0^\infty e^{-y} dy + \int_0^{-y} ye^{-y} dy \right]$$

$$f_X(x) = \frac{1}{2}e^{-x}(x+1)$$

Marginal PDF of Y:

By symmetry, the marginal PDF is:

$$f_Y(y) = \frac{1}{2}e^{-y}(y+1)$$

Exercise 5.

$$f_{X,Y}(x,y) = \begin{cases} ke^{-(x+y)}, & x \geq 0, y \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

(a)

1. Non-negativity

The given function $f_{X,Y}(x,y) = ke^{-(x+y)}$ is non-negative for all $x \geq 0$ and $y \geq 0$ because:

- the constant k is non-negative
- the exponential function $e^{-(x+y)}$ is always positive for $x \geq 0$ and $y \geq 0$. Therefore, the non-negativity condition is satisfied.

2. Normalization

$$\int_0^\infty \int_0^\infty ke^{-(x+y)} dx dy = 1$$

$$k \int_0^\infty e^{-y} \int_0^\infty e^{-x} dx dy = 1$$

$$k \int_0^\infty e^{-y} \times 1 dy = 1$$

$$k \times 1 = 1$$

$$k = 1$$

To satisfy the normalization condition $k = 1$, therefore, the joint PDF is: $f_{X,Y}(x,y) = e^{-(x+y)}$ for $x \geq 0$ and $y \geq 0$

(b) $P(X < 1 \text{ and } Y < 1) = \int_0^1 \int_0^1 f_{X,Y}(x,y) dx dy$

$$= \int_0^1 \int_0^1 e^{-(x+y)} dx dy$$

$$\begin{aligned}
&= \int_0^1 e^{-y} dy \int_0^1 e^{-x} dx \\
&= [-e^{-y}|_0^1] [-e^{-x}|_0^1] \\
&= [-e^{-1} + 1] [-e^{-1} + 1] \\
&= (1 - \frac{1}{e})^2
\end{aligned}$$

$$(c) P(X + Y > 3)$$

$$P(X + Y > 3) = 1 - P(X + Y \leq 3)$$

$$\begin{aligned}
P(X + Y \leq 3) &= \int \int_{x+y \leq 3} f_{X,Y}(x, y) dx dy \\
&= \int \int_{x+y \leq 3} e^{-(x+y)} dx dy
\end{aligned}$$

$$\begin{aligned}
P(X + Y \leq 3) &= \int_0^3 \int_0^u e^{-u} dv du \\
&= \int_0^3 [ve^{-u}|_0^u] du \\
&= \int_0^3 ue^{-u} du
\end{aligned}$$

$$a = u, db = e^{-u} du$$

$$da = du, b = -e^{-u}$$

$$\begin{aligned}
&= -ue^{-u} + \int_0^3 e^{-u} du \\
&= -ue^{-u} - e^u|_0^3 \\
&= -(u + 1)e^{-u}|_0^3 \\
&= -4e^{-3} + e^0 \\
&= 1 - 4e^{-3}
\end{aligned}$$

$$\begin{aligned}
P(X + Y > 3) &= 1 - (1 - 4e^{-3}) \\
&= 4e^{-3}
\end{aligned}$$

Exercise 6.

Let X be the time it takes Alana to solve problems, and Y be the time it takes Beatrice to solve the problem. Both X and Y are independent and exponentially distributed with parameter λ .

$$f_X(x) = \lambda e^{-\lambda x} \text{ for } x \geq 0$$

$$f_Y(y) = \lambda e^{-\lambda y} \text{ for } y \geq 0$$

$$\begin{aligned} P(X \geq 2Y) &= \int_0^\infty \lambda e^{-\lambda y} \int_{2y}^\infty \lambda e^{-\lambda x} dx dy \\ &= \lambda^2 \int_0^\infty e^{-\lambda y} \int_{2y}^\infty e^{-\lambda x} dx dy \\ &= \lambda^2 \int_0^\infty e^{-\lambda y} \left[-\frac{1}{\lambda} e^{-\lambda x} \Big|_{2y}^\infty \right] dy \\ &= \lambda^2 \int_0^\infty e^{-\lambda y} \left[\frac{1}{\lambda} e^{-\lambda 2y} + 0 \right] dy \\ &= \lambda \int_0^\infty e^{-3\lambda y} dy \end{aligned}$$

$$u = 3\lambda y$$

$$du = 3\lambda dy$$

$$\begin{aligned} &= \frac{1}{3} \int_0^\infty e^{-u} du \\ &= \frac{1}{3} \times 1 \\ &= \frac{1}{3} \end{aligned}$$

Exercise 7.

$$X \sim \text{Exp}\left(\frac{3}{2}\right)$$

$$Y \sim N(0, 1)$$

$$P(X > Y^2) = \int_{-\infty}^\infty P(X > Y^2 | Y = y) f_Y(y) dy$$

$$X \sim \text{Exp}\left(\frac{3}{2}\right) : f_X(x) = \frac{3}{2} e^{-\frac{3}{2}x}, x \geq 0$$

$$P(X > Y^2 | Y = y) = \int_{y^2}^\infty \frac{3}{2} e^{-\frac{3}{2}x} dx = -e^{-\frac{3}{2}x} \Big|_{y^2}^\infty = e^{-\frac{3}{2}y^2}$$

$$\begin{aligned} P(X > Y^2) &= \int_{-\infty}^\infty e^{-\frac{3}{2}y^2} \times \frac{1}{\sqrt{2\pi}} e^{-\frac{y^2}{2}} dy \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^\infty e^{-2y^2} dy \\ &= \frac{1}{\sqrt{2\pi}} \sqrt{\frac{\pi}{2}} \\ &= \frac{\sqrt{\pi}}{\sqrt{2}\sqrt{\pi}\sqrt{2}} \end{aligned}$$

$$= \frac{1}{2}$$

Exercise 8.

$$X \sim \text{unif}(0, 1)$$

$$Y \sim \text{unif}(0, 1)$$

$$P(|X - Y| \leq 0.25)$$

$$-0.25 \leq X - Y \leq 0.25$$

Since X and Y are uniformly distributed on the interval $(0, 1)$, their joint distribution over the unit square $[0, 1] \times [0, 1]$ has a constant joint density of 1 in this region. Therefore, the probability $P(|X - Y| \leq 0.25)$ can be calculated as the area of the region within $[0, 1] \times [0, 1]$ where $|X - Y| \leq 0.25$

$$\begin{cases} X - Y = -0.25 \\ X - Y = 0.25 \end{cases}$$

$$\begin{cases} X = -0.25 + Y \\ X = 0.25 + Y \end{cases}$$

$$\begin{cases} Y = X + 0.25 \\ Y = X - 0.25 \end{cases}$$

$$\mathbf{Y = X + 0.25}$$

$$(0, 0.25) \text{ and } (0.75, 1)$$

$$\mathbf{Y = X - 0.25}$$

$$(0.25, 0) \text{ and } (1, 0.75)$$

$$\text{Area} = 1 - \frac{1}{2}((0.75 - 0)(1 - 0.25)) - \frac{1}{2}((0.75 - 0)(1 - 0.25))$$

$$= 1 - \frac{1}{2}(0.75 \times 0.75) - \frac{1}{2}(0.75 \times 0.75)$$

$$= 1 - \frac{1}{2} \times 0.5625 - \frac{1}{2} \times 0.5625$$

$$= 1 - 0.28125 - 0.28125$$

$$= 0.4375$$

Exercise 9.

$$P(X > Y) = \int \int_{(x,y):x>y} f(x)f(y) dx dy$$

$$P(X > Y) = \int_{-\infty}^{\infty} \int_{-\infty}^y f(x)f(y) dx dy$$

$$P(X > Y) = \int_{-\infty}^{\infty} f(y) \int_{-\infty}^y f(x) dx dy$$

$$P(X > Y) = \int_{-\infty}^{\infty} f(y)(1 - F(y)) dy \text{ or } P(X > Y) = \int_{-\infty}^{\infty} f(y)F(y) dy$$

$$P(X > Y) = \int_{-\infty}^{\infty} f(y) dy - \int_{-\infty}^{\infty} f(y)F(y) dy = \int_{-\infty}^{\infty} dy$$

$$P(X > Y) = 1 - \int_{-\infty}^{\infty} f(y)F(y) dy = \int_{-\infty}^{\infty} f(y)F(y) dy$$

$$1 = 2 \int_{-\infty}^{\infty} f(y)F(y) dy$$

$$\frac{1}{2} = \int_{-\infty}^{\infty} f(y)F(y) dy$$

$$P(X > Y) = 1 - \frac{1}{2} = \frac{1}{2}$$

Exercise 10.

$$X - \text{Gamma}(2, 1)$$

$$Y - \text{Unif}(0, 1)$$

$$f_X(x) = xe^{-x}$$

$$f_Y(y) = 1$$

$$P(XY \leq u) = P(Y \leq \frac{u}{x}) = \int_0^1 P(X \leq \frac{u}{y}) f_Y(y) dy$$

$$= \int_0^1 F_X(\frac{u}{y}) \times 1 dy$$

$$F_X(\frac{u}{y}) = \int_0^{\frac{u}{y}} xe^{-x} dx$$

$$u = x, dv = e^{-x}$$

$$du = 1, v = -e^{-x}$$

$$F_X(\frac{u}{y}) = -xe^{-x} + \int_0^{\frac{u}{y}} e^{-x} dx$$

$$F_X\left(\frac{u}{y}\right) = [-xe^{-x} - e^{-x}]_0^{\frac{u}{y}}$$

$$F_X\left(\frac{u}{y}\right) = -\frac{u}{y}e^{-\frac{u}{y}} - e^{-\frac{u}{y}}$$

$$\begin{aligned} &= \int_0^1 -\frac{u}{y}e^{-\frac{u}{y}} - e^{-\frac{u}{y}} dy \\ &= \frac{d}{du} \left(\int_0^1 -\frac{u}{y}e^{-\frac{u}{y}} - e^{-\frac{u}{y}} dy \right) \\ &= \int_0^1 \left(\frac{d}{du} -\frac{u}{y}e^{-\frac{u}{y}} - e^{-\frac{u}{y}} \right) dy \\ &= \int_0^1 \left(-\frac{1}{y}e^{-\frac{u}{y}} - \frac{u}{y^2}e^{-\frac{u}{y}} + \frac{1}{y}e^{-\frac{u}{y}} \right) dy \\ &= \int_0^1 -\frac{u}{y^2}e^{-\frac{u}{y}} dy \end{aligned}$$

$$v = \frac{u}{y}, y = \frac{u}{v}, dy = -\frac{u}{v^2} dv$$

$$\begin{aligned} &= -u \int_{\infty}^u \frac{1}{\left(\frac{u}{v}\right)^2} e^{-v} \times \left(-\frac{u}{v^2}\right) dv \\ &= u \int_u^{\infty} \frac{v^2}{u^2} \times \frac{u}{v^2} e^{-v} dv \\ &= u \int_u^{\infty} \frac{1}{u} e^{-v} dv \\ &= \int_u^{\infty} e^{-v} dv \\ &= e^{-v} \Big|_u^{\infty} \\ &= e^{-u}, u \geq 0 \end{aligned}$$

Therefore, $U = XY - \text{Exp}(1)$

Exercise 11.

- X is selected uniformly at random from $(0, \frac{L}{2})$
- Y is selected uniformly at random from $(\frac{L}{2}, L)$

$$P(|Y - X| > \frac{L}{3}) =$$

X is uniformly distributed over $(0, \frac{L}{2})$, so its PDF is: $f_X(x) = \frac{1}{\frac{L}{2}-0} = \frac{1}{\frac{L}{2}} = \frac{2}{L}$ for $x \in (0, \frac{L}{2})$

Y is uniformly distributed over $(\frac{L}{2}, L)$, so its PDF is: $f_Y(y) = \frac{1}{L-\frac{L}{2}} = \frac{1}{\frac{L}{2}} = \frac{2}{L}$ for $y \in (\frac{L}{2}, L)$

$$\begin{aligned}
P(Y - X > \frac{L}{3}) &= \int_0^{\frac{L}{6}} \int_{\frac{L}{2}}^L f_X(x) f_Y(y) dy dx + \int_{\frac{L}{6}}^{\frac{L}{3}} \int_{x+\frac{1}{3}}^L f_X(x) f_Y(y) dy dx \\
&= \int_0^{\frac{L}{6}} \int_{\frac{L}{2}}^L \frac{2}{L} \times \frac{2}{L} dy dx + \int_{\frac{L}{6}}^{\frac{L}{3}} \int_{x+\frac{1}{3}}^L \frac{2}{L} \times \frac{2}{L} dy dx \\
&= \frac{4}{L^2} [\int_0^{\frac{L}{6}} \int_{\frac{L}{2}}^L dy dx + \int_{\frac{L}{6}}^{\frac{L}{3}} \int_{x+\frac{1}{3}}^L dy dx] \\
&= \frac{4}{L^2} [\int_0^{\frac{L}{6}} [y]_{\frac{L}{2}}^L dx + \int_{\frac{L}{6}}^{\frac{L}{3}} [y]_{x+\frac{1}{3}}^L dx] \\
&= \frac{4}{L^2} [\int_0^{\frac{L}{6}} \frac{L}{2} dx + \int_{\frac{L}{6}}^{\frac{L}{3}} \frac{2L}{3} - x dx] \\
&= \frac{4}{L^2} ([\frac{L}{2}x]_0^{\frac{L}{6}} + [\frac{2L}{3}x - \frac{1}{2}x^2]_{\frac{L}{6}}^{\frac{L}{3}}) \\
&= \frac{4}{L^2} ([\frac{L^2}{12}] + [\frac{2L^2}{6} - \frac{L^2}{8} - \frac{2L^2}{18} + \frac{L^2}{72}]) \\
&= \frac{4}{L^2} (\frac{L^2}{12} + \frac{L^2}{9}) \\
&= \frac{4}{L^2} (\frac{7L^2}{36}) \\
&= \frac{7}{9}
\end{aligned}$$

Exercise 12.

$$U - f_U(u) = \begin{cases} \frac{u}{2}, & 0 < u < 2 \\ 0, & \text{otherwise} \end{cases}$$

$$V - f_V(v) = \begin{cases} 2v, & 0 < v < 1 \\ 0, & \text{otherwise} \end{cases}$$

$$\tan(\theta) = \frac{U}{V}$$

$$P(\tan(\theta) < \tan(\frac{\pi}{4})) = P(\frac{U}{V} < 1) = P(U < V)$$

$$\begin{aligned}
P(U < V) &= \int_0^1 \int_0^v f_U(u) f_V(v) du dv \\
&= \int_0^1 \int_0^v 2v \times \frac{u}{2} du dv \\
&= \int_0^1 v [\frac{1}{2}u^2]_0^v dv \\
&= \frac{1}{2} \int_0^1 v \times v^2 dv \\
&= \frac{1}{2} \int_0^1 v^3 dv
\end{aligned}$$

$$\begin{aligned} &= \frac{1}{8} v^4 \Big|_0^1 \\ &= \frac{1}{8} \end{aligned}$$